NeuroAgent

Building and validating an evidence-driven neurological Agentic System

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Motivation

Motivation — Dataset — Agent — Collaboration — Supervision

A new way of testing medical AI

Most research today tests medical AI with question answering. Real clinical work is not like that.

Today: question answering

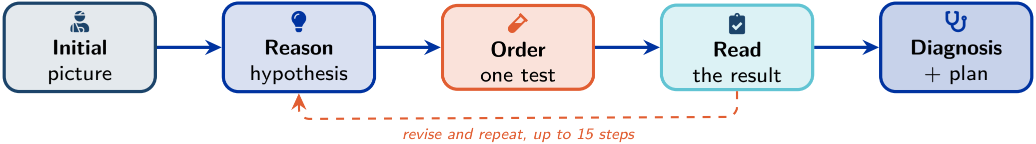
- Multiple-choice exams
- Every clue is already in the question
- One correct option to pick
- Rewards recall, not reasoning

Real clinical practice

- The patient arrives with a complaint, not options
- You decide which investigations to order
- Results come back over time
- Cost, risk and safety all matter

We introduce a **new paradigm**: test the AI on the whole diagnostic workup, the way a clinician actually reasons.

Project overview



600 realistic neurology cases across 20 conditions.

What we measure

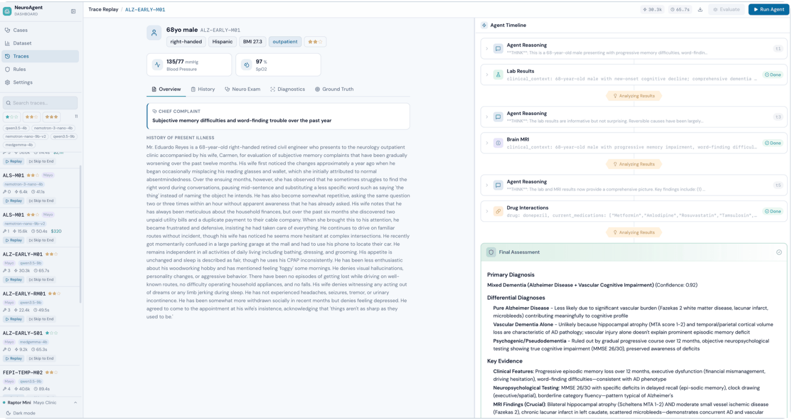
- Right test, right order, sensible cost
- Sound reasoning, not memorised facts
- A workup a clinician can follow

Why it is new

- Today's AI exams hand over every clue
- Tests the real skill: what to investigate
- Each case checked by **2+ clinicians**

The agent in action

The goal: a transparent workup, ending in a diagnosis backed by evidence.



Dataset

Motivation **Dataset** Agent Collaboration Supervision

The dataset

Cases

600

20 conditions

Per condition

30

exactly, fully balanced

Follow-up results

3,800

triggered by actions

How each case is built

- Patient core: history, exam, vitals
- Evidence hidden behind tool calls
- Ground truth: best path, red herrings, unsafe actions

Spanning

- 3 difficulty tiers: straightforward to puzzle
- 396 synthetic + 204 real-report-seeded
- 3 settings: emergency / inpatient / outpatient

*Results are cached in two tiers: **initial** findings when a test is first ordered, and **follow-ups** unlocked by a specific next action.*

How the cases were generated

Two pipelines, one schema: real cases for fidelity, synthetic cases for scale.

Seeded from real case reports

- Source: **MedCaseReasoning**, clinician-written case reports
- Rewritten by **Claude Opus 4.8 + Sonnet 4.6**: findings hidden in tool outputs

204 cases

Fully synthetic, for scale

- Built from per-condition clinical schemas, no source report
- Scale needed to **fine-tune local models**
- Same schema and ground truth as the seeded cases

396 cases

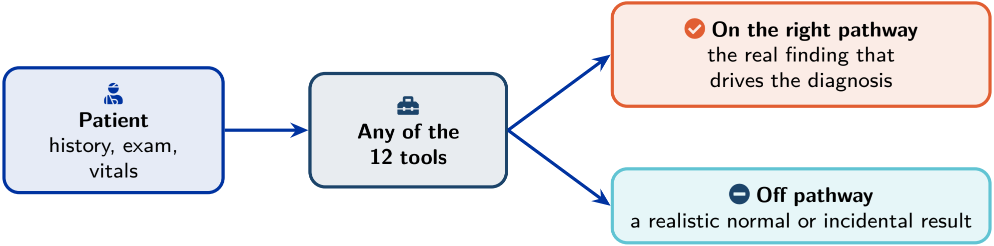
The 12 diagnostic tools

Tool	What it returns
Brain MRI	report with region-tagged findings
EEG	background, epileptiform features, focality
ECG	rhythm, intervals, signs of ischaemia
Labs	panels with reference ranges + abnormality flags
CSF (lumbar puncture)	cells, protein, glucose, NfL, oligoclonal bands
CT scan	non-contrast / contrast, head / spine
Echocardiogram	chambers, valves, ejection fraction, thrombus
Cardiac monitoring	telemetry / Holter / loop recorder
Advanced imaging	MRA / MRV / CTA / PET / DaT-SPECT
Specialised tests	EMG/NCS, biopsy, autoimmune panels
Literature search	guideline and paper snippets
Drug interactions	interaction profile + contraindications

Each returns a structured, typed report with a cost; any tool, any time.

Every patient is a complete picture

Each case pre-computes a whole patient, so the agent can order **any** of the 12 tools and always get a realistic answer.



Never a dead end: every tool answers. But each call costs money and a turn, so ordering the wrong test is penalised.

A worked example: wake-up ischemic stroke

Turn 1: what the agent sees

- 58 F: woke with slurred speech, clumsy hand
- Last known well 9 h ago: window unclear
- Left face/arm weak, sensory loss, neglect

What the right workup reveals

- CT first: rule out bleed; glucose: no hypoglycemia
- MRI: fresh infarct, deep right white matter
- ECG, echo, carotids, Holter: hunt the cause

The trap, and an off-pathway test

"Chronic, not acute" tempts an early stop, yet the cause hunt is still owed. An off-pathway EEG reads **normal**: wasted.

Ground truth: acute ischemic stroke, right corona radiata (ICD-10 I63.541).

Agent

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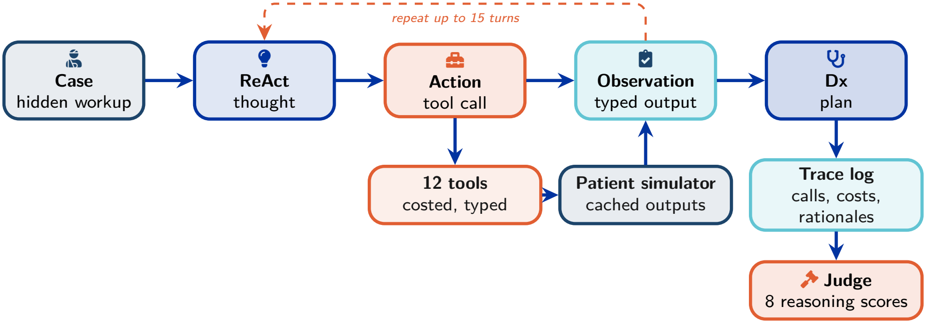
Agent stack: from patient to judged trace



Why this matters clinically

Every run is replayable: ordered tests, revealed evidence, cost/safety events, and an independent reasoning-judge score.

Preliminary base-model results

Model	Composite	Top-1	Top-3	Tool F1	Safety fails
qwen3.5-9b	0.779	35.3%	63.3%	0.56	28/150
qwen3.5-4b	0.671	36.7%	48.7%	0.46	45/150
nemotron-nano-9b-v2	0.565	11.3%	24.0%	0.28	82/150
gemma-4-e2b	0.538	13.3%	33.3%	0.35	72/150
gemma-4-e4b	0.483	12.0%	19.3%	0.26	90/150
nemotron-3-nano-4b	0.339	5.3%	8.0%	0.06	130/150

What the pilot says

50 random v5 cases, 3 repeats per case and model: 900 ReAct traces judged by an independent LLM judge. Qwen3.5 leads on composite score and top-3 calibration.

Failure modes to validate

Lumbar puncture ordered before imaging, unsafe medication advice, text-only fake tool calls, and weak reasoning to positively diagnose functional neurological disorder recur across models.

What those metrics mean

Rule-based checks

- **Top-1:** final diagnosis matches the ground-truth ICD group
- **Top-3:** correct ICD group appears anywhere in the differential
- **Tool F1:** called tools vs the optimal workup sequence
- **Safety fails:** contraindicated, harmful, or wrong-order actions

LLM-judge composite

- **Judge dims:** 8 clinical-reasoning scores, each 0–5
- **Composite:** weighted mean, normalised to 0–1
- **Main weights:** diagnosis accuracy + clinical safety
- **Also checks:** evidence use, differential, uncertainty, tool efficiency, red herrings

Clinician validation: cases and sampled runs, including field correctness, hypothesis quality, unsafe reasoning, and score/clinical-quality alignment.

ICD = International Classification of Diseases, the WHO standard diagnostic coding system used to check whether the model's diagnosis matches the ground truth.

Fine-tuning approach: SFT w golden trajectories

Supervised Fine Tuning corpus

1. **Input:** clinician-approved optimal action sequence
2. **Trace:** rationale → tool call → cached result → next rationale
3. **Filter:** correct diagnosis, no unsafe / wrong-order actions
4. **Lock:** train only on clinician-validated cases

LoRA / QLoRA

LoRA: adapters; base frozen
QLoRA: 4-bit frozen base
Trainable: 116.4M / 9.07B params (1.28%)
Fits: long ReAct traces on one A100-40 GB

Hybrid model: attention LoRA only on the 8/32 full-attention layers; MLP LoRA on all 32.

Why SFT is not enough

Learns: ideal path
Misses: recovery after a wrong early test
Needs: scored sampled trajectories

RL roadmap: optimise scored trajectories

GRPO *(planned)*

- **Sample:** several workups for one case
- **Score:** benchmark each trajectory
- **Update:** improve vs group average
- **No value model**

DAPO *(planned)*

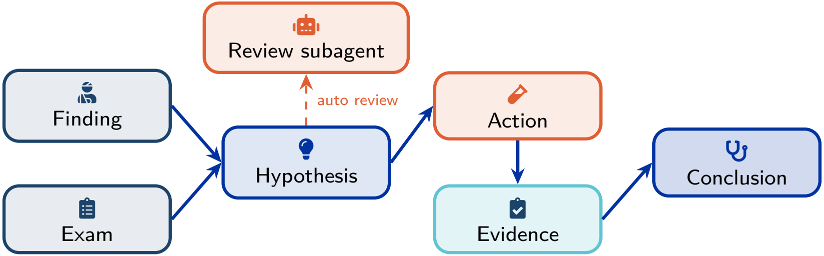
- **Target:** longer reasoning traces
- **Keep:** useful hard trajectories
- **Reduce:** noisy clipping effects
- **Focus:** updates where gains remain

Reward signal

trajectory → benchmark scores → composite reward → policy update

diagnosis | evidence | tools/order | cost | contraindications | red herrings | uncertainty

Hypothesis DAG: make reasoning inspectable



Typed nodes

Finding / exam: observed facts
Hypothesis: candidate diagnosis
Action / evidence: test and result
Conclusion: final diagnosis + plan

Typed edges

`supports`: adds weight
`contradicts`: weakens a hypothesis
`triggers`: justifies an action
`depends-on`: records prerequisites

System-triggered hypothesis review subagent

New hypothesis $\xrightarrow{\text{automatic}}$ Reviewer → Critique → DAG writeback

Trigger rule

When: each new hypothesis node
Who: system, not acting agent
Why: critique cannot be skipped

Review checklist

differentials | evidence gaps | contradictions
premature closure | unsafe next actions
test justification

Structured writeback

`severity` | `claim` | `evidence_node`
`recommended_fix` | `clinician_review_needed`

Clinician validation

cases + sampled runs
hypothesis quality + unsafe reasoning
clinical usefulness of DAG critiques

Collaboration

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Goals for the medical collaboration

1. Validate the cases

- Read each patient as you would in clinic
- Approve, flag, or correct field by field
- Drop cases that are not good enough

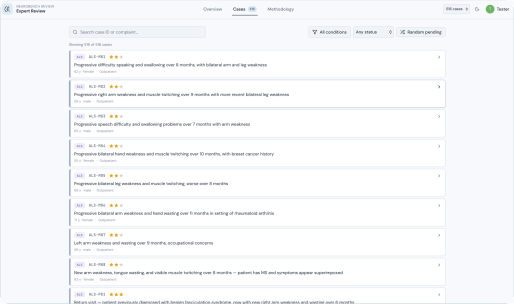
2. Validate our scoring

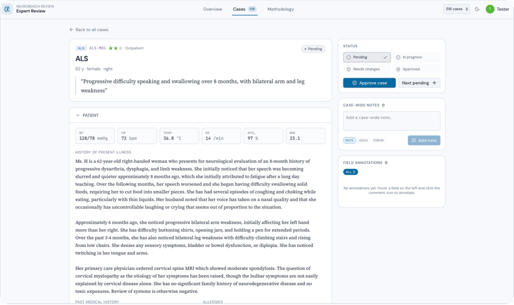
- Score a sample of agent runs yourself
- Compare against the AI reasoning judge
- Confirm scores track clinical quality

How the review works

- $\geq$**2 independent reviewers** per case
- Separate login per reviewer: blind, no shared bias
- Agreement stats $\rightarrow$ fix or drop weak cases $\rightarrow$ higher quality

The annotation platform

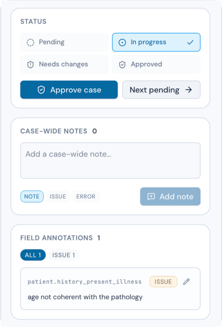




- Browse cases: condition, difficulty, status
- One screen per patient: vitals, history, full workup

review.andreaprotani.com

Reviewing a case



Flag any field: note, issue, or error

Per-case status, every annotation logged

Approve, request changes, or move on: progress tracked per reviewer.

Supervision

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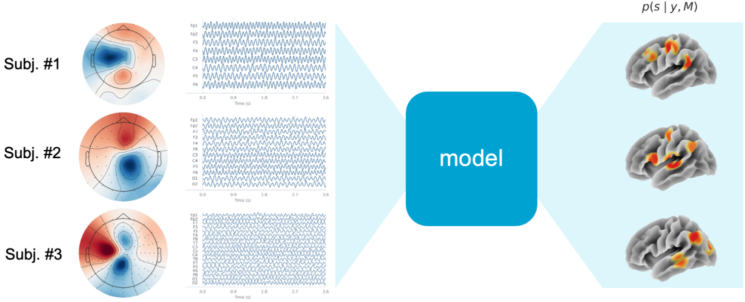
Research objective

Goal: EEG source localization → estimate a distribution over plausible cortical sources from EEG, while remaining **montage-agnostic**.

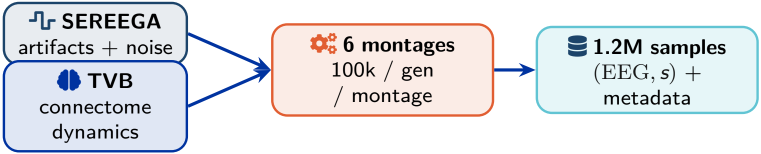
Related MSc thesis project, supervised with **Lorenzo Sibi** | linkedin.com/in/lorenzo-sibi



The output should expose **uncertainty**: more than one source map can fit the same EEG.

Synthetic dataset generation

Training needs many paired (EEG, source) samples; real ground truth needs intracerebral stimulation, which is rare.



Artifacts & noise simulated

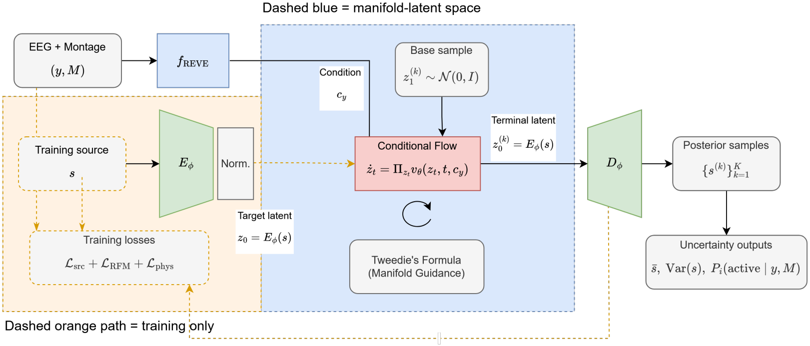
- Eye-blink / muscle artifacts, line noise, sensor drift
- Pink/white noise; connectivity-driven background (TVB)
- SEREEGA | TVB

Why synthetic data

- Need large paired sets; ground truth needs real stimulation
- Only **Localize-MI** has it – real-world test set
- Mikulan et al., 2020, Sci. Data

Draft architecture

A conditional flow samples source latents from EEG and montage context, then decodes them into posterior cortical maps.



Training signal
source labels + physics-aware losses

Posterior sampling
multiple plausible source maps

Outputs
mean, variance, active probability

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